

Base-three prime digit sums cannot be uniform modulo two

CONJECTURE 00000000793 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Digit congruence

In base 3, an integer and its digit sum have the same parity because $3 \equiv 1 \pmod{2}$. Every prime other than 2 is odd. Thus only the prime 2 has an even base-three digit sum.

Limit contradiction

Among the first N primes the even-digit-sum frequency is between 0 and $1/N$. It therefore tends to 0, instead of the uniform-modulo-two value $1/2$. Base 3 is a prime base, so this single base refutes the universal assertion.

Scope

The filed statement concerns all residues modulo $p-1$. A statement restricted to admissible unit residues is not refuted by this example.

Formalization

Lean proves the actual digit-sum congruence, classifies the even-digit-sum primes, uses the standard n th-prime enumeration, bounds the empirical frequency by $1/N$, and proves its limit is zero and cannot be one-half.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000793.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.